Fourth-order interpolation: production validation

Branch `atom-fourier-interpolation`; implementation `e98eafcc`

Verdict. The fourth-order interpolation passes the finite-range polynomial benchmark. No defect was found in its resolved-support operator. A separate bug in `ignore_v3` was found and fixed: it previously removed only one direction of the cubic coupling. The combined regression suite passes all 56 tests, including 16 new deterministic production tests.

Controlled physical model. Let x_n be a mass-weighted molecular internal displacement, $s_n(h) = \sum_r h_r x_{n+r}$, and $\omega_q^2 = 1 + 0.8\cos(2\pi q)$. In units $\hbar = m = a = 1$, the potential is

$$V = \tfrac{1}{2}x^T K x + \frac{g}{6}\sum_n (s_n^3 - 3C_h s_n) + \sum_{d,n} \frac{\lambda_d}{24}(s_{d,n}^4 - 6C_d s_{d,n}^2 + 3C_d^2). \tag{1}$$

The cubic coefficients are $(g, h) = (0.08, [1, 0.35])$; the quartics are $(\lambda, h) = (0.12, [1, 0.37, -0.29, 0.21])$ and $(0.04, [1, -0.23])$. Positive quartics bound the potential below. The cutoff is three cells, and the first quartic couples *four distinct cells*. At $Q = 0$, the optical frequency $\sqrt{1.8}$ lies above the two-phonon threshold $2\sqrt{0.2}$.

$C_h = \langle s(h)^2 \rangle$ is fixed from the prescribed coarse Gaussian. The counterterms cancel its mean anharmonic force and curvature. **Neither the potential nor the harmonic reference is re-minimized on the fine mesh.** Degree-seven Gaussian quadrature exactly integrates the polynomial force estimator, eliminating Monte Carlo uncertainty.

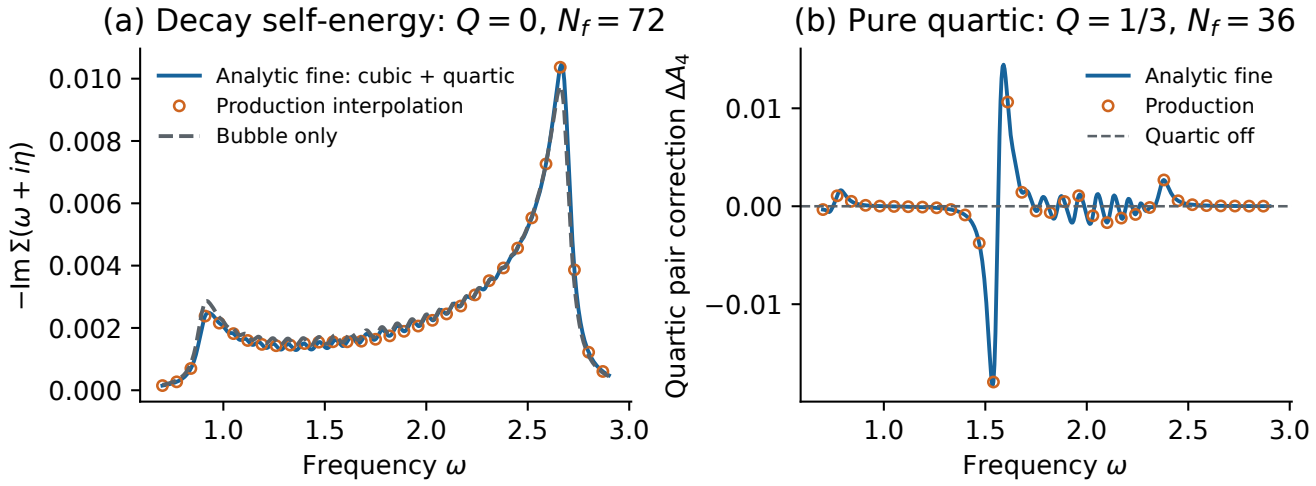


Figure 1: Actual production fold–Julia contraction–reconstruction, compared with analytic fine-mesh derivatives of the same potential. (a) Imaginary self-energy, with and without quartic resummation, at $T = 0$. (b) Quartic change in the pair response, with cubic coupling disabled, at $k_B T = 0.3$: the difference of $-\mathrm{Im}\,\mathrm{Tr}(z^2 I + L_{\mathrm{pair}})^{-1}/(\pi d)$ with and without the quartic vertex; d is the pair-block dimension. Both use $N_c = 9$ and $z = \omega + 0.04i$. Open markers overlay the analytic curves; dashed curves show that the quartic contribution is measurable. This is a regulated finite-mesh response, not an extrapolated thermodynamic lifetime.

Why the test establishes interpolation accuracy. The independent reference uses analytic vertices $v = \Phi^{(3)}/\sqrt{N_f}$ and $W = \Phi^{(4)}/N_f$, with the full ordered-pair convention:

$$\Sigma(z) = v^\dagger[I - \Lambda(z)W]^{-1}\Lambda(z)v, \qquad G(z) = [z^2 - \omega_Q^2 - \Sigma(z)]^{-1}. \tag{2}$$

At fixed external Q, a finite-range quartic vertex is a trigonometric polynomial in *two independent internal momenta*. Production interpolates both pair legs with normalization N_c/N_f. When the real-space support is resolved by the centered coarse mesh, the tensor is reconstructed before response inversion. The reference does not reuse that interpolation.

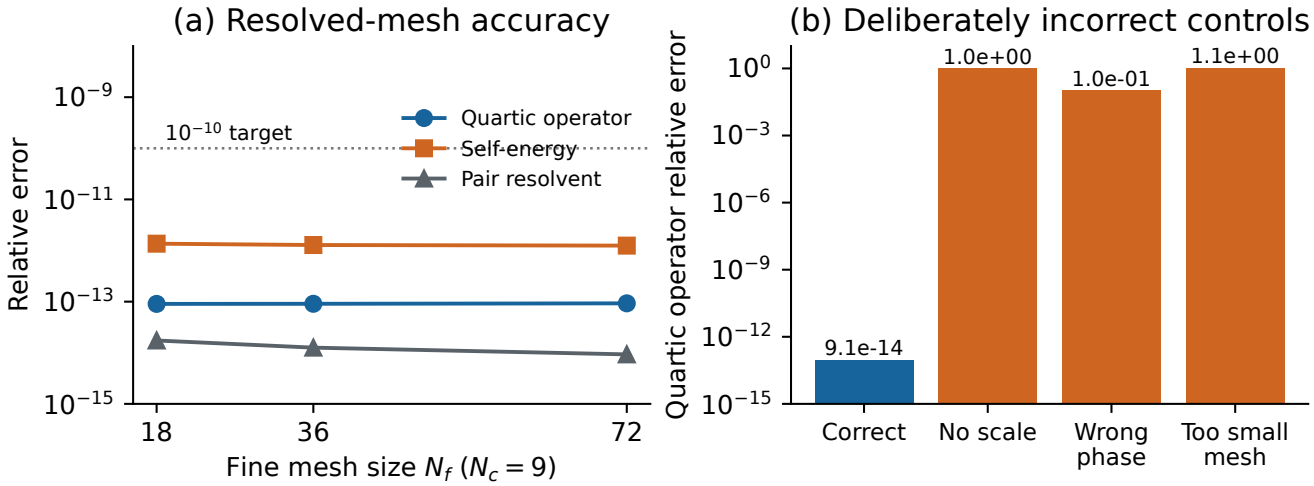


Figure 2: (a) Production versus independent analytic fine reference at $Q = 1/3$, $k_BT = 0.3$, and $\eta = 0.04$. Operator errors are relative Frobenius norms after subtracting the harmonic block; self-energy and pair-resolvent errors are maxima over 41 frequencies in $[0.3, 2.8]$. (b) Negative controls: omitting the mesh factor, conjugating the kernel incorrectly, or using $N_c = 3$ for the same three-cell interaction produces large quartic errors. The first three bars use $9 \to 18$, $Q = 1/3$; the last uses $3 \to 18$, $Q = 0$; all four controls have $T = 0$.

The regression checks isolate cubic and quartic operators, bubble and resummed self-energies, and the pure-quartic pair resolvent. They include $9 \to 9/18/36/72$ and $8 \to 24$ meshes, nonzero Q, finite-temperature sum/difference channels, unequal masses, atom rebasing, dense complex mode mixing, and acoustic masks. Vertices and resolvents satisfy 10^{-10} tolerances; self-energy satisfies 10^{-8}.

Interpolating an *already computed response* is a different operation: the response is not finite range. The separate analytic off-grid control at $Q = 1/18$ gives about 31% error, even using identical fine internal quadrature. Production currently requires external Q on the coarse mesh; this control does not claim otherwise.

What was wrong and what was right. The quartic interpolation was right in these resolved-support tests. The `ignore_v3` switch was wrong: it removed the one-phonon-to-pair coupling but retained the reverse coupling. Both q-space classes now remove both directions, restoring pure-quartic isolation. This fix does not change the usual calculation with cubic coupling enabled.

Reproduction. From this checkout with its Python/Julia environment, run `benchmarks/interpolation/make_report_figures.py`. The plotted arrays and source hashes accompany the vector figures. The exact test command is in the adjacent `README.md`. Compile this file with `pdflatex` from its directory.